FIBONACCI CHARACTER SUMS AT RANK INDEX TWO AND JACOBSTHAL SUMS

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ABSTRACT. Let $p > 5$ be prime, let $z(p)$ be the rank of apparition of p in the Fibonacci sequence, and put

$$S(p) = \sum_{j=1}^{z(p)} \left(\frac{F_j}{p}\right), \qquad i_p = \frac{p - \left(\frac{5}{p}\right)}{z(p)}.$$

Earlier work of the author evaluates $S(p)$ in the full-rank case $i_p = 1$. This paper treats the next layer, namely the rank-index-two case $i_p = 2$.

We first show that $i_p = 2$ forces $p \equiv 1 \pmod 4$. In the $p \equiv 1 \pmod 8$ branches, a change of variables through a primitive root in the split case and a Cayley parametrisation of the norm-one torus in the inert case reduce $S(p)$ to classical Jacobsthal sums; the complementary $p \equiv 5 \pmod 8$ branches are handled by reflection and character identities.

Writing

$$\mathcal{J}(c) = \sum_{x \in \mathbb{F}_p} \left(\frac{x}{p}\right)\left(\frac{x^2 + c}{p}\right),$$

we obtain the following explicit laws. In the split case, with

$$\delta_p = \begin{cases} 1, & p \equiv 1 \pmod 5, \\ -1, & p \equiv 4 \pmod 5, \end{cases}$$

and with $p = a^2 + b^2$, a odd, $b > 0$ even, and $a^* \in \{\pm a\}$ determined by $a^* \equiv 1 \pmod 4$, one has

$$S(p) = \begin{cases} \delta_p a^*, & p \equiv 1 \pmod 8, \\ -\delta_p, & p \equiv 5 \pmod 8. \end{cases}$$

In the inert case one has

$$S(p) = \begin{cases} \frac{1}{2}\mathcal{J}(5) \in \{\pm b\}, & p \equiv 1 \pmod 8, \\ 0, & p \equiv 5 \pmod 8, \end{cases}$$

and in the first branch the sign is determined by the congruence $S(p) \equiv -a^* \pmod 5$.

A companion verifier checks the theorem-level identities and records the numerical data used in the paper; the source code and machine-readable outputs are available in the versioned reproducibility archive [6].

1. INTRODUCTION

1.1. **Setting.** For a prime $p > 5$ let $\chi = \chi_p = \left(\frac{\cdot}{p}\right)$, $\varepsilon_p = \left(\frac{5}{p}\right)$, $n_p = p - \varepsilon_p$, let $z(p)$ be the rank of apparition of p in (F_n), and

$$i_p := \frac{n_p}{z(p)} \in \mathbb{Z}_{\geq 1}, \qquad S(p) := \sum_{j=1}^{z(p)} \chi(F_j).$$

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Earlier work [5] — the *full-rank paper* — gives an explicit evaluation of $S(p)$ for $i_p = 1$; for split primes it reads $S(p) = \delta_p$, where $\delta_p \in \{\pm 1\}$ and $\delta_p \equiv p \pmod 5$. The density of the primes with $i_p = 1$ is, under GRH, a special case of Chen's theorem on primitive points of one-dimensional tori [2]: the total density is Artin's constant A, with inert and split branches $10A/19$ and $9A/19$ respectively (see the conditional corollary in [5]). The maximal-order problem for units in real quadratic fields was initiated numerically by Ishikawa–Kitaoka [8] and settled conditionally by Chen–Kitaoka–Yu [3] and Roskam [20]; for the fundamental unit φ (rather than the norm-one element $\eta = -\varphi^2$), Roskam's inert value is $20A/19$. See Lenstra [14] and Murty [17] for the general conditional framework. The present note asks what happens one level down, at $i_p = 2$.

1.2. **Main results.** Write $p = a^2 + b^2$ with a odd, b even and $b > 0$ whenever $p \equiv 1 \pmod 4$, and let $a^* \in \{\pm a\}$ be determined by $a^* \equiv 1 \pmod 4$. Let $\mathcal{J}(c) = \sum_{x \in \mathbb{F}_p} \chi(x)\chi(x^2 + c)$ be the Jacobsthal sum.

Theorem A (Split primes, index two)**.** *Let p be split ($\varepsilon_p = 1$) with $i_p = 2$. Then $p \equiv 1 \pmod 4$, and*

$$S(p) = \begin{cases} \delta_p\, a^*, & p \equiv 1 \pmod 8, \\ -\delta_p, & p \equiv 5 \pmod 8. \end{cases}$$

Theorem B (Inert primes, index two)**.** *Let p be inert, that is $\varepsilon_p = \left(\frac{5}{p}\right) = -1$, and suppose $i_p = 2$. Then $p \equiv 1 \pmod 4$. If $p \equiv 5 \pmod 8$, then*

$$S(p) = 0.$$

If $p \equiv 1 \pmod 8$, write

$$p = (a^*)^2 + b^2, \qquad a^* \equiv 1 \pmod 4, \qquad b > 0.$$

Then

$$S(p) = \frac{1}{2}\mathcal{J}(5) = \frac{1}{2}\mathcal{J}(-5),$$

and the sign is given by

$$S(p) = \begin{cases} -b, & b \equiv a^* \pmod 5, \\ \ \ b, & b \equiv -a^* \pmod 5. \end{cases}$$

Equivalently, when $p \equiv 1 \pmod 8$, $S(p)$ is the unique element of $\{\pm b\}$ congruent to $-a^*$ modulo 5.

Proposition 1.1 (Reflection vanishing)**.** *If $z(p)$ is odd, then $p \equiv 1 \pmod 4$. Moreover, if in addition $p \equiv 5 \pmod 8$, then $S(p) = 0$.*

1.3. **What is classical and what is claimed.**

Ingredient	Source	Role here
i_p even $\iff p \equiv 1 \pmod 4$	classical (Lehmer [12])	Lemma 3.2
Jacobsthal: $(\frac{1}{2}\mathcal{J}(r))^2 + (\frac{1}{2}\mathcal{J}(n))^2 = p$	Jacobsthal [9], [1]	input
$\mathcal{J}(-1) = \sum \chi(x^3 - x) = -2a^*$ ($p \equiv 1 \bmod 8$)	Gauss; [1, Ch. 6,9]	input
$\left(\frac{5}{p}\right)_4 \chi(\varphi) = \delta_p$	elementary lemma; proof supplied here	Lemma 3.7
Rank multiplier / reflection	classical (Lucas [16], Somer [21]); see also [4]	Lemmas 3.3, 3.4
Half-Jacobsthal reduction ($p \equiv 1 \pmod 8$, $i_p = 2$)	this paper	Props. 4.1, 5.2
Sign of $\frac{1}{2}\mathcal{J}(5)$ via biquadratic reciprocity	classical reciprocity + application here	Lemma 5.4, Thm. B
Inert index-two Dirichlet density $= 10A/19$ (GRH)	Roskam [20] + Chen [2] + 2-adic dictionary (§7)	Cor. 7.2
Theorems A, B	this paper	main results
Proposition 1.1	classical; also used in [4]	structural

The contribution claimed here is the reduction of the Fibonacci character sum at rank index two to the specific Jacobsthal sums $\mathcal{J}(-1)$ and $\mathcal{J}(\pm 5)$, together with the resulting explicit formulas for $S(p)$. The classical evaluation theory of Jacobsthal sums and the relevant reciprocity laws are used as inputs; no new general evaluation theory for Jacobsthal sums is claimed. We have not located a prior reference recording these particular rank-index-two Fibonacci reductions, nor their combination with the quartic-reciprocity sign determination in the inert branch. This is a qualified literature statement, not a proof of priority. The inert Dirichlet density $10A/19$ recorded in Corollary 7.2 below is derived from the GRH-conditional density theorems of Roskam and Chen after translating their statements into the present rank-index notation; it is included only as conditional context (Remark 7.3).

1.4. **Organisation.** Section 2 fixes notation. Section 3 proves the structural lemmas. Sections 4 and 5 prove Theorems A and B. Section 6 records the numerical verification of the theorem-level identities (including the signed inert law). Section 7 records a GRH-conditional density corollary obtained from Roskam's and Chen's density theorems after translating their maximal-order statements into the present rank-index notation. This conditional discussion is included only for context and is not used in the proofs of Theorems A and B. Section 8 records further problems and directions.

2. Notation and preliminaries

Let $\varphi = (1+\sqrt{5})/2$, $\psi = (1-\sqrt{5})/2$, so $\varphi\psi = -1$ and $F_n = (\varphi^n - \psi^n)/(\varphi - \psi)$. Reducing modulo p we regard $\varphi, \psi \in \mathbb{F}_p$ if $\varepsilon_p = 1$ and $\varphi, \psi \in \mathbb{F}_{p^2} \setminus \mathbb{F}_p$ with $\varphi^p = \psi$ if $\varepsilon_p = -1$. Put

$$\eta := \varphi/\psi = -\varphi^2.$$

Then $F_n \equiv 0 \pmod p \iff \eta^n = 1$, so $z(p) = \operatorname{ord}(\eta)$ and $\eta^{n_p} = 1$.

Definition 2.1. For $p \equiv 1 \pmod 4$ fix $\iota \in \mathbb{F}_p$ with $\iota^2 = -1$; then $\chi(\iota) = (-1)^{(p-1)/4} = \left(\frac{2}{p}\right)$. For split $p \equiv 1 \pmod 4$ put $\left(\frac{5}{p}\right)_4 := \chi(\varphi - \psi) \equiv 5^{(p-1)/4} \pmod p$; this does not depend on the choice of square root since $\chi(-1) = 1$. For $p \equiv 1 \pmod 8$ write $2 = s^2$ in $\mathbb{F}_p$ and set $\left(\frac{2}{p}\right)_4 := \chi(s)$; this is the quartic symbol of 2.

Definition 2.2 (Jacobsthal sums)**.** $\mathcal{J}(c) := \sum_{x\in\mathbb{F}_p} \chi(x)\chi(x^2+c)$ for $c \in \mathbb{F}_p^\times$.

Remark 2.3. Throughout this paper, the term "Jacobsthal sum" refers to the classical quadratic character sum

$$\mathcal{J}(c) = \sum_{x\in\mathbb{F}_p} \chi(x)\chi(x^2+c).$$

It is unrelated to the Jacobsthal and Jacobsthal–Lucas recurrence sequences commonly denoted by J_n and j_n.

Lemma 2.4 (Jacobsthal)**.** *Let $p \equiv 1 \pmod 4$, let $r \in \mathbb{F}_p^\times$ be a quadratic residue and let $n \in \mathbb{F}_p^\times$ be a quadratic non-residue. Then*

$$\frac{1}{2}\mathcal{J}(r) \text{ is odd}, \qquad \frac{1}{2}\mathcal{J}(n) \text{ is even},$$

and

$$\left(\frac{1}{2}\mathcal{J}(r)\right)^2 + \left(\frac{1}{2}\mathcal{J}(n)\right)^2 = p.$$

Moreover, for every $c, u \in \mathbb{F}_p^\times$,

$$\mathcal{J}(cu^2) = \chi(u)\mathcal{J}(c).$$

Proof. The first assertions are the classical theorem of Jacobsthal on the two components in the representation of primes $p \equiv 1 \pmod 4$ as a sum of two squares; see Jacobsthal [9] or Berndt–Evans–Williams [1, Chapter 6].

For the transformation formula, substitute $x = uy$:

$$\begin{aligned}
\mathcal{J}(cu^2) &= \sum_{x\in\mathbb{F}_p} \chi(x)\chi(x^2+cu^2) \\
&= \sum_{y\in\mathbb{F}_p} \chi(uy)\chi(u^2(y^2+c)) \\
&= \chi(u)\sum_{y\in\mathbb{F}_p} \chi(y)\chi(y^2+c) \\
&= \chi(u)\mathcal{J}(c).
\end{aligned}$$

□

Lemma 2.5 (Jacobsthal congruence)**.** *Let $p \equiv 1 \pmod 4$, and write*

$$p = a^2 + b^2, \qquad a \text{ odd}, \quad b \text{ even},$$

with $a^* \in \{\pm a\}$ determined by $a^* \equiv 1 \pmod 4$. Let $c \in \mathbb{F}_p^\times$, and put $m = (p-1)/2$. Then

$$\mathcal{J}(c) \equiv -\binom{m}{m/2} c^{(p-1)/4} \pmod p.$$

By the classical Gauss congruence

$$\binom{m}{m/2} \equiv 2a^* \pmod p,$$

one obtains

$$\tfrac{1}{2}\mathcal{J}(c) \equiv -a^*\, c^{(p-1)/4} \pmod p.$$

In particular, for $p \equiv 1 \pmod 8$,

$$\mathcal{J}(-1) = -2a^*.$$

Proof. Working in $\mathbb{F}_p$, one has $\chi(x) \equiv x^m$ for all $x \in \mathbb{F}_p$, with both sides equal to 0 at $x = 0$. Hence

$$\mathcal{J}(c) = \sum_{x\in\mathbb{F}_p} \chi(x)\chi(x^2+c) \equiv \sum_{x\in\mathbb{F}_p} x^m(x^2+c)^m \pmod{p}.$$

Expanding the binomial gives

$$\sum_{k=0}^{m} \binom{m}{k} c^{m-k} \sum_{x\in\mathbb{F}_p} x^{m+2k}.$$

Using the rule

$$\sum_{x\in\mathbb{F}_p} x^e \equiv \begin{cases} -1 \pmod{p}, & (p-1) \mid e,\ e > 0, \\ 0 \pmod{p}, & \text{otherwise}, \end{cases}$$

only the middle term $k = m/2$ survives. Therefore

$$\mathcal{J}(c) \equiv -\binom{m}{m/2} c^{m/2} = -\binom{m}{m/2} c^{(p-1)/4} \pmod{p}.$$

The evaluation of the central binomial coefficient is classical:

$$\binom{m}{m/2} \equiv 2a^* \pmod{p};$$

see, for example, Berndt–Evans–Williams [1, Chapters 6 and 9]. Thus

$$\tfrac{1}{2}\mathcal{J}(c) \equiv -a^*\, c^{(p-1)/4} \pmod{p}.$$

Now suppose $p \equiv 1 \pmod 8$ and put $c = -1$. Then

$$(-1)^{(p-1)/4} = 1,$$

so

$$\tfrac{1}{2}\mathcal{J}(-1) \equiv -a^* \pmod{p}.$$

Since -1 is a quadratic residue, Lemma 2.4 implies that $\frac{1}{2}\mathcal{J}(-1)$ is the odd component in the representation $p = a^2 + b^2$. Hence

$$\tfrac{1}{2}\mathcal{J}(-1) = \pm a^*.$$

The congruence modulo p determines the sign. Indeed, $2|a^*| < p$ for $p > 5$, so a^* and $-a^*$ are not congruent modulo p. Therefore

$$\tfrac{1}{2}\mathcal{J}(-1) = -a^*,$$

or equivalently

$$\mathcal{J}(-1) = -2a^*.$$

□

Lemma 2.6. *Let $p \equiv 1 \pmod 8$, and write*

$$p = a^2 + b^2, \qquad a \text{ odd}, \quad b \text{ even}, \quad b > 0,$$

with $a^* \in \{\pm a\}$ determined by $a^* \equiv 1 \pmod 4$. Then

$$\mathcal{J}(-1) = \sum_{x\in\mathbb{F}_p} \chi(x^3 - x) = -2a^*.$$

Proof. This is the final assertion of Lemma 2.5. □

Remark 2.7 (Uniform evaluation of $\mathcal{J}(-1)$)**.** Lemma 2.6 records the classical evaluation for the branch $p \equiv 1 \pmod 8$. The same ingredients yield a uniform formula for every $p \equiv 1 \pmod 4$:

$$\mathcal{J}(-1) = -2\left(\frac{2}{p}\right) a^*.$$

Indeed, Lemma 2.5 gives $\frac{1}{2}\mathcal{J}(-1) \equiv -a^*(-1)^{(p-1)/4} \pmod p$, while Lemma 2.4 identifies $\frac{1}{2}\mathcal{J}(-1)$ with the odd component $\pm a^*$ of $p = a^2 + b^2$. The congruence selects the sign in both residue classes modulo 8. When $p \equiv 1 \pmod 8$ one has $\left(\frac{2}{p}\right) = 1$, recovering $\mathcal{J}(-1) = -2a^*$; when $p \equiv 5 \pmod 8$ one has $\left(\frac{2}{p}\right) = -1$ and $\mathcal{J}(-1) = 2a^*$. The companion verifier checks the uniform identity on all index-two records with $p \equiv 1 \pmod 4$ in the archived range (zero mismatches).

3. Structural lemmas

Lemma 3.1 (Parity)**.** $S(p) \equiv z(p) - 1 \pmod 2$.

Proof. For $1 \le j < z(p)$ one has $F_j \not\equiv 0$, so $\chi(F_j) = \pm 1$; and $\chi(F_{z(p)}) = 0$. □

Lemma 3.2 (Index parity)**.** $\eta^{n_p/2} = \chi(-1)$. *Hence i_p is even if and only if $p \equiv 1 \pmod 4$.*

Proof. If $\varepsilon_p = 1$: $\eta^{(p-1)/2} = (-1)^{(p-1)/2}\varphi^{p-1} = (-1)^{(p-1)/2}$. If $\varepsilon_p = -1$: $\varphi^{p+1} = \varphi\psi = -1$, so $\eta^{(p+1)/2} = (-1)^{(p+1)/2}\varphi^{p+1} = (-1)^{(p-1)/2}$. Finally i_p is even iff $z(p) \mid n_p/2$ iff $\eta^{n_p/2} = 1$. □

Lemma 3.3 (Rank multiplier)**.** *Let $z = z(p)$. Then $F_{z+1} = \varphi^z \in \mathbb{F}_p$ and $F_{z+1}^2 = (-1)^z$. Consequently $\chi(F_{z+1}) = 1$ if z is even, while if z is odd then $p \equiv 1 \pmod 4$ and $\chi(F_{z+1}) = \left(\frac{2}{p}\right)$.*

Proof. $F_z = 0$ means $\varphi^z = \psi^z = (-1)^z\varphi^{-z}$, so $\varphi^{2z} = (-1)^z$, and $F_{z+1} = (\varphi^{z+1} - \psi^{z+1})/(\varphi - \psi) = \varphi^z$. If z is odd, $F_{z+1}^2 = -1$ forces $p \equiv 1 \pmod 4$ and $F_{z+1} = \pm\iota$. □

Lemma 3.4 (Reflection)**.** *For $0 \le j \le z = z(p)$: $F_{z-j} = (-1)^{j+1}F_{z+1}F_j$.*

Proof. Using $\psi^z = \varphi^z$ and $\varphi\psi = -1$: $\varphi^{z-j} - \psi^{z-j} = \varphi^z(\varphi^{-j} - \psi^{-j}) = \varphi^z(-1)^j(\psi^j - \varphi^j)$; divide by $\varphi - \psi$ and apply Lemma 3.3. □

Proof of Proposition 1.1. Let $z = z(p)$ be odd. Since p is odd and

$$n_p = p - \varepsilon_p$$

is even, the quotient

$$i_p = \frac{n_p}{z}$$

is even. Hence Lemma 3.2 gives $p \equiv 1 \pmod 4$, so $\chi(-1) = 1$. By Lemma 3.4, $j \mapsto z - j$ is a fixed-point-free involution of $\{1, \dots, z-1\}$ with $\chi(F_{z-j}) = \chi(F_{z+1})\chi(F_j)$ (note that the factor $\chi((-1)^{j+1})$ is absorbed because $\chi(-1) = 1$), and $\chi(F_z) = 0$. Hence $S(p) = \chi(F_{z+1})S(p)$, and Lemma 3.3 gives $\chi(F_{z+1}) = \left(\frac{2}{p}\right)$. In particular $S(p) = 0$ whenever $\left(\frac{2}{p}\right) = -1$, i.e., $p \equiv 5 \pmod 8$. □

Corollary 3.5. *$S(p) = 0$ in each of the following cases: (a) $\varepsilon_p = -1$, $i_p = 2$, $p \equiv 5 \pmod 8$; (b) $\varepsilon_p = 1$, $i_p = 4$, $p \equiv 5 \pmod 8$.*

Proof. In (a) $z = (p+1)/2$ and in (b) $z = (p-1)/4$; both are odd. □

Lemma 3.6. *If $\varepsilon_p = 1$ and $i_p = 2$, then $\chi(\varphi) = -\left(\frac{2}{p}\right)$.*

Proof. Here $p \equiv 1 \pmod 4$ and $\operatorname{ord}\eta = (p-1)/2$ is even, so $\eta^{(p-1)/4} = -1$. But $\eta^{(p-1)/4} = (-1)^{(p-1)/4}\varphi^{(p-1)/2} = \left(\frac{2}{p}\right)\chi(\varphi)$. □

Lemma 3.7 (Quartic character of the golden unit)**.** *Let $p > 5$ be a split prime for $K = \mathbb{Q}(\sqrt{5})$, and suppose $p \equiv 1 \pmod 4$. Define*

$$\delta_p = \begin{cases} 1, & p \equiv 1 \pmod 5, \\ -1, & p \equiv 4 \pmod 5. \end{cases}$$

Then

$$\left(\frac{5}{p}\right)_4 \chi(\varphi) = \delta_p.$$

Equivalently,

$$\chi(\varphi + 2) = \delta_p.$$

Proof. Since

$$\varphi(\varphi - \psi) = \varphi\sqrt{5} = \varphi + 2,$$

we have

$$\left(\frac{5}{p}\right)_4 \chi(\varphi) = \chi(\sqrt{5})\chi(\varphi) = \chi(\varphi + 2).$$

Thus it suffices to determine whether $\varphi + 2$ is a square in $\mathbb{F}_p$.

The element $\varphi - 1$ is a root of the quadratic

$$X^2 + X - 1 = 0.$$

Let $\zeta \in \mathbb{F}_{p^2}$ be a primitive fifth root of unity. Since $p \equiv \pm 1 \pmod 5$, Frobenius acts by

$$\zeta \longmapsto \zeta^p \in \{\zeta, \zeta^{-1}\}.$$

Hence $\zeta + \zeta^{-1} \in \mathbb{F}_p$, and the two elements

$$\zeta + \zeta^{-1}, \qquad \zeta^2 + \zeta^{-2}$$

are precisely the roots $\varphi - 1, \psi - 1$ of $X^2 + X - 1$. Replacing ζ by ζ^2 if necessary, we may assume that

$$\zeta + \zeta^{-1} = \varphi - 1.$$

Put

$$\theta = \zeta - \zeta^{-1}.$$

Then

$$\theta^2 = (\zeta + \zeta^{-1})^2 - 4 = (\varphi - 1)^2 - 4 = -(\varphi + 2).$$

Because $p \equiv 1 \pmod 4$, -1 is a square in $\mathbb{F}_p$. Hence $\varphi + 2$ is a square in $\mathbb{F}_p$ if and only if θ^2 is a square in $\mathbb{F}_p$.

If $p \equiv 1 \pmod 5$, then $\zeta \in \mathbb{F}_p$, hence $\theta \in \mathbb{F}_p$, and therefore θ^2, and so also $\varphi + 2$, is a square.

If $p \equiv 4 \pmod 5$, then Frobenius acts on ζ by

$$\zeta^p = \zeta^{-1}.$$

Consequently

$$\theta^p = (\zeta - \zeta^{-1})^p = \zeta^{-1} - \zeta = -\theta.$$

Thus $\theta \notin \mathbb{F}_p$. Since the only square roots of θ^2 are $\pm\theta$, the element θ^2 cannot be a square in $\mathbb{F}_p$. Because -1 is a square, $\varphi + 2 = -\theta^2$ is also a nonsquare.

Therefore

$$\chi(\varphi + 2) = \begin{cases} 1, & p \equiv 1 \pmod 5, \\ -1, & p \equiv 4 \pmod 5, \end{cases}$$

which is precisely δ_p. □

4. The split case

Proposition 4.1 (Jacobsthal reduction, split)**.** *Let $\varepsilon_p = 1$ and $i_p = 2$. Then*

$$S(p) = \begin{cases} \left(\frac{5}{p}\right)_4 \cdot \frac{1}{2}\mathcal{J}(-1), & p \equiv 1 \pmod 8, \\ -\left(\frac{5}{p}\right)_4, & p \equiv 5 \pmod 8. \end{cases}$$

Proof. By Lemma 3.2, $p \equiv 1 \pmod 4$, and $z = (p-1)/2$.

Step 1. Put $g = \iota\varphi$ and $y_j = g^j$. Since $\psi^j = (-1)^j\varphi^{-j}$, $F_j = \varphi^{-j}\big(\varphi^{2j} - (-1)^j\big)/(\varphi - \psi)$. Now $y_j^2 = (-1)^j\varphi^{2j}$, so $\varphi^{2j} - (-1)^j = (-1)^j(y_j^2 - 1)$, and $\chi(\varphi^{-j}) = \chi(\varphi^j) = \chi(\iota)^j\chi(y_j)$. Therefore

$$\chi(F_j) = \left(\frac{5}{p}\right)_4 \left(\frac{2}{p}\right)^j f(y_j), \qquad f(y) := \chi(y)\chi(y^2 - 1).$$

Step 2. Let $d = \mathrm{ord}_p(g)$. The identity $\mathrm{ord}_p(g^2) = d/\gcd(d, 2)$, together with $\mathrm{ord}_p(g^2) = (p-1)/2$, shows $d \mid p-1$ and $d \in \{(p-1)/2,\ p-1\}$. If $d = (p-1)/2$, then — since $(p-1)/2$ is even — $\mathrm{ord}_p(g^2) = d/2 = (p-1)/4 \neq (p-1)/2$, a contradiction. Hence $d = p-1$, so g is a primitive root. Consequently $\{y_j : 1 \le j \le z\}$ is a half-system of $\mathbb{F}_p^\times$ modulo ± 1, because $g^{j+z} = -g^j$. Also f is even, since $\chi(-1) = 1$.

Step 3. If $p \equiv 1 \pmod 8$, then $\left(\frac{2}{p}\right) = 1$, so $S(p) = \left(\frac{5}{p}\right)_4 \sum_{j=1}^{z} f(y_j) = \left(\frac{5}{p}\right)_4 \cdot \frac{1}{2}\sum_{y\neq 0} f(y) = \left(\frac{5}{p}\right)_4 \cdot \frac{1}{2}\mathcal{J}(-1)$.

If $p \equiv 5 \pmod 8$, then $\left(\frac{2}{p}\right)^j = (-1)^j = \chi(g^j)$, so $S(p) = \left(\frac{5}{p}\right)_4 \sum_j \chi(y_j^2 - 1) = \left(\frac{5}{p}\right)_4 \cdot \frac{1}{2}\sum_{y\neq 0}\chi(y^2 - 1)$. Since

$$\sum_{y\in\mathbb{F}_p} \chi(y^2 - 1) = -1$$

and $\chi(-1) = 1$, we have

$$\sum_{y\neq 0} \chi(y^2 - 1) = -2.$$

Consequently

$$S(p) = \left(\frac{5}{p}\right)_4 \cdot \frac{1}{2}(-2) = -\left(\frac{5}{p}\right)_4.$$

□

Proof of Theorem A. By Lemma 3.7,

$$\left(\frac{5}{p}\right)_4 \chi(\varphi) = \delta_p.$$

Since $\chi(\varphi)^2 = 1$, this gives

$$\left(\frac{5}{p}\right)_4 = \delta_p\chi(\varphi).$$

By Lemma 3.6,

$$\chi(\varphi) = -\left(\frac{2}{p}\right).$$

Therefore

$$\left(\frac{5}{p}\right)_4 = -\delta_p\left(\frac{2}{p}\right).$$

If $p \equiv 1 \pmod 8$, then $\left(\frac{2}{p}\right) = 1$, so $\left(\frac{5}{p}\right)_4 = -\delta_p$. By Proposition 4.1 and Lemma 2.6,

$$S(p) = \left(\frac{5}{p}\right)_4 \cdot \frac{1}{2}\mathcal{J}(-1) = (-\delta_p)(-a^*) = \delta_p a^*.$$

If $p \equiv 5 \pmod 8$, then $\left(\frac{2}{p}\right) = -1$, so $\left(\frac{5}{p}\right)_4 = \delta_p$. By Proposition 4.1,

$$S(p) = -\left(\frac{5}{p}\right)_4 = -\delta_p.$$

□

Corollary 4.2. *If $\varepsilon_p = 1$ and $i_p = 2$, then $\left(\frac{5}{p}\right)_4 = -\delta_p \left(\frac{2}{p}\right)$. Equivalently, 5 is a quartic residue modulo such p if and only if $p \equiv 9$ or $21 \pmod{40}$.*

Proof. The first identity was established in the proof of Theorem A; the residue classes follow by inspection of the four cases p mod 40. □

Remark 4.3 (Compatibility with a criterion of E. Lehmer)**.** Emma Lehmer [13] proved that for primes $p \equiv 1 \pmod 4$ with $\left(\frac{5}{p}\right) = 1$ one has $\left(\frac{5}{p}\right)_4 = 1$ if and only if $5 \mid b$. Combined with Corollary 4.2 this yields the consistency check that, for split primes with $i_p = 2$,

$$5 \mid b \quad \Longleftrightarrow \quad p \equiv 9 \text{ or } 21 \pmod{40}.$$

The equivalence has been verified by the companion verifier on every split record of the archived run below $2 \cdot 10^6$ (23,190 primes with $i_p \in \{2, 4\}$), in particular on all 13,134 split primes with $i_p = 2$.

Remark 4.4 (Parity consistency)**.** For split $i_p = 2$, $z(p) = (p-1)/2$ is even, so $S(p)$ is odd by Lemma 3.1, in accordance with the values $\delta_p a^*$ and $-\delta_p$. For inert $i_p = 2$, $z(p)$ is odd, so $S(p)$ is even, in accordance with $\pm b$ and 0.

Remark 4.5. In terms of residue classes: $S(p) = a^*$ for $p \equiv 1 \pmod{40}$, $S(p) = -a^*$ for $p \equiv 9 \pmod{40}$, $S(p) = -1$ for $p \equiv 21 \pmod{40}$, and $S(p) = 1$ for $p \equiv 29 \pmod{40}$. Compare the full-rank law $S(p) = \delta_p$ of [5]. For $i_p = 2$ one has

$$S(p) \equiv \delta_p \left(\frac{2}{p}\right) \equiv -\left(\frac{5}{p}\right)_4 \pmod 4.$$

A uniform statement covering all indices remains open.

5. The inert case

Let $\varepsilon_p = -1$. Let $N : \mathbb{F}_{p^2}^\times \to \mathbb{F}_p^\times$ be the norm and $T = \ker N$, a cyclic group of order $p+1$. Put $\sqrt{5} := \varphi - \psi$; then $(\sqrt{5})^p = -\sqrt{5}$.

Lemma 5.1 (Cayley parametrisation)**.** *The map $c(t) = (t+\sqrt{5})/(t-\sqrt{5})$, $c(\infty) = 1$, is a bijection $\mathbb{P}^1(\mathbb{F}_p) \to T$ with $c(0) = -1$ and $c(5/t) = -c(t)$. For $t \in \mathbb{F}_p^\times$*

$$\frac{c(t) - c(t)^{-1}}{\sqrt{5}} = \frac{4t}{t^2 - 5} \in \mathbb{F}_p^\times.$$

Proof. Standard: $c(t)^p = c(t)^{-1}$, and the inverse is $y \mapsto \sqrt{5}\,(y+1)/(y-1)$, which is Frobenius-fixed. Since 5 is a non-residue, $t^2 \neq 5$. The identities follow by direct computation. □

Proposition 5.2 (Jacobsthal reduction, inert)**.** *Let $\varepsilon_p = -1$ and $i_p = 2$. If $p \equiv 1 \pmod 8$, then*

$$S(p) = \frac{1}{2}\mathcal{J}(-5) = \frac{1}{2}\mathcal{J}(5).$$

Proof. Here $p \equiv 1 \pmod 4$ by Lemma 3.2, and

$$z = z(p) = \frac{p+1}{2}$$

is odd. Put

$$g = \iota\varphi.$$

Then

$$N(g) = N(\iota)N(\varphi) = (-1)(-1) = 1,$$

so $g \in T$, and

$$g^2 = \eta$$

has order z.

With $y_j = g^j$ one has the identities

$$(-\iota)^j y_j = \varphi^j, \qquad (-\iota)^j y_j^{-1} = (-1)^j \varphi^{-j} = \psi^j,$$

and therefore

$$F_j = (-\iota)^j \, \frac{y_j - y_j^{-1}}{\sqrt{5}}.$$

(For $y \in T$, the quotient $(y - y^{-1})/\sqrt{5}$ lies in $\mathbb{F}_p$, because Frobenius sends $y - y^{-1}$ to $-(y - y^{-1})$ and sends $\sqrt{5}$ to $-\sqrt{5}$.) Consequently

$$\chi(F_j) = \left(\frac{2}{p}\right)^j \chi\left(\frac{y_j - y_j^{-1}}{\sqrt{5}}\right).$$

Since $p \equiv 1 \pmod 8$, we have $\left(\frac{2}{p}\right) = 1$, so this becomes

$$\chi(F_j) = \chi\left(\frac{y_j - y_j^{-1}}{\sqrt{5}}\right).$$

We next record explicitly how the powers of g sample the norm-one torus. Since $g^2 = \eta$ has order $z = (p+1)/2$, the order of g is either z or $2z = p+1$. In both cases the set

$$Y = \{g^j : 1 \le j \le z\}$$

contains exactly one element from each pair $\{y, -y\}$ in T.

Indeed, if $\mathrm{ord}(g) = 2z$, then $g^z = -1$, and therefore

$$g^{j+z} = -g^j.$$

Thus the first z powers give one representative from each pair $\{y, -y\}$.

If instead $\mathrm{ord}(g) = z$, then $\langle g \rangle$ is the unique subgroup of T of order z. Since z is odd, this subgroup does not contain -1. Hence

$$T = \langle g \rangle \sqcup (-\langle g \rangle),$$

and again $Y = \langle g \rangle$ contains exactly one element from each pair $\{y, -y\}$.

Now use the Cayley parametrisation of Lemma 5.1. Write

$$y = c(t) = \frac{t + \sqrt{5}}{t - \sqrt{5}}.$$

For $t \in \mathbb{F}_p^\times$, Lemma 5.1 gives

$$\frac{c(t) - c(t)^{-1}}{\sqrt{5}} = \frac{4t}{t^2 - 5}.$$

Therefore

$$\chi\left(\frac{c(t)-c(t)^{-1}}{\sqrt{5}}\right)=\chi(t)\chi(t^2-5).$$

Put

$$f(t):=\chi(t)\chi(t^2-5).$$

The involution $y\mapsto -y$ on T corresponds, under the Cayley parametrisation, to

$$t\mapsto \frac{5}{t}.$$

(Alternatively, without Cayley: $(-y)-(-y)^{-1}=-(y-y^{-1})$, so the character is invariant under $y\mapsto -y$ as soon as $\chi(-1)=1$, i.e., already under the weaker hypothesis $p\equiv 1 \pmod 4$.) Moreover,

$$f\left(\frac{5}{t}\right)=f(t).$$

Indeed, a direct computation yields

$$\begin{aligned}
f\left(\frac{5}{t}\right)&=\chi\left(\frac{5}{t}\right)\chi\left(\frac{25}{t^2}-5\right)\\
&=\chi(5)\chi(t)\,\chi\left(\frac{5(5-t^2)}{t^2}\right)\\
&=\chi(5)^2\chi(t)\chi(5-t^2)\\
&=\chi(t)\chi(-(t^2-5))\\
&=\chi(t)\chi(t^2-5)\\
&=f(t),
\end{aligned}$$

where the last step uses only $\chi(-1)=1$ (i.e., $p\equiv 1 \pmod 4$).

The pair $\{1,-1\}\subset T$ corresponds to $t=\infty$ and $t=0$. Both give zero contribution: for $y=1$ one has $y-y^{-1}=0$, and for $y=-1$ one again has $y-y^{-1}=0$. Hence summing over one representative from each pair $\{y,-y\}$ is the same as taking one half of the sum over all nonzero finite parameters $t\in\mathbb{F}_p^\times$. Therefore

$$S(p)=\frac{1}{2}\sum_{t\in\mathbb{F}_p^\times}\chi(t)\chi(t^2-5).$$

The term $t=0$ is zero, so

$$S(p)=\frac{1}{2}\sum_{t\in\mathbb{F}_p}\chi(t)\chi(t^2-5)=\frac{1}{2}\mathcal{J}(-5).$$

Finally, substituting $x=\iota u$ gives

$$\begin{aligned}
\mathcal{J}(-5)&=\sum_{x\in\mathbb{F}_p}\chi(x)\chi(x^2-5)\\
&=\sum_{u\in\mathbb{F}_p}\chi(\iota u)\chi(-u^2-5)\\
&=\chi(\iota)\chi(-1)\sum_{u\in\mathbb{F}_p}\chi(u)\chi(u^2+5)\\
&=\left(\frac{2}{p}\right)\mathcal{J}(5).
\end{aligned}$$

Since $p \equiv 1 \pmod 8$, $\left(\frac{2}{p}\right) = 1$, and hence

$$\mathcal{J}(-5) = \mathcal{J}(5).$$

This proves the proposition. □

Definition 5.3 (Quartic residue symbol in $\mathbb{Z}[i]$)**.** Let π be a primary Gaussian prime of odd norm

$$N(\pi) = p,$$

and let $\alpha \in \mathbb{Z}[i]$ be coprime to π. The quartic residue symbol

$$\left(\frac{\alpha}{\pi}\right)_4 \in \{1, -1, i, -i\}$$

is the unique fourth root of unity satisfying the Euler criterion

$$\alpha^{(p-1)/4} \equiv \left(\frac{\alpha}{\pi}\right)_4 \pmod \pi.$$

An odd Gaussian integer is called primary if it is congruent to 1 modulo $2 + 2i$. We use the standard quartic reciprocity law for primary Gaussian integers; see Ireland–Rosen [7, Chapter 9, §§8–9, pp. 121–126].

Lemma 5.4 (Sign via biquadratic reciprocity)**.** *Let $p \equiv 1 \pmod 8$ with $\left(\frac{5}{p}\right) = -1$, and write*

$$p = (a^*)^2 + b^2, \qquad a^* \equiv 1 \pmod 4, \qquad b > 0.$$

Put

$$\pi = a^* + bi.$$

With the convention of Definition 5.3, one has $5 \nmid a^* b$,

$$b \equiv \pm a^* \pmod 5,$$

and

$$\left(\frac{5}{\pi}\right)_4 = \begin{cases} i, & b \equiv a^* \pmod 5, \\ -i, & b \equiv -a^* \pmod 5. \end{cases}$$

Proof. We use Definition 5.3 throughout. Since $\left(\frac{5}{p}\right) = -1$, one has

$$p \equiv 2 \text{ or } 3 \pmod 5.$$

The quadratic residues modulo 5 are $0, 1, 4$. From

$$p = (a^*)^2 + b^2$$

it follows that neither a^* nor b is divisible by 5, and the two squares must be equal modulo 5: they are both 1 if $p \equiv 2 \pmod 5$, and both 4 if $p \equiv 3 \pmod 5$. Hence

$$b^2 \equiv (a^*)^2 \pmod 5,$$

and therefore

$$r := \frac{b}{a^*} \equiv \pm 1 \pmod 5.$$

Write

$$5 = \lambda\bar{\lambda}, \qquad \lambda = -1 + 2i, \qquad \bar{\lambda} = -1 - 2i.$$

Both λ and $\bar{\lambda}$ are primary. Also π is primary: since $p \equiv 1 \pmod 8$, one has $b \equiv 0 \pmod 4$ and $a^* \equiv 1 \pmod 4$, so $\pi \equiv 1 \pmod{2 + 2i}$.

For relatively prime primary Gaussian primes α and β, quartic reciprocity gives

$$\left(\frac{\alpha}{\beta}\right)_4 = (-1)^{\frac{N(\alpha)-1}{4}\frac{N(\beta)-1}{4}} \left(\frac{\beta}{\alpha}\right)_4;$$

see Ireland–Rosen [7, Chapter 9, §9, pp. 123–126]. Since

$$N(\lambda) = N(\bar{\lambda}) = 5$$

and

$$\frac{p-1}{4}$$

is even, the sign factor is trivial. Therefore

$$\left(\frac{\lambda}{\pi}\right)_4 = \left(\frac{\pi}{\lambda}\right)_4, \qquad \left(\frac{\bar{\lambda}}{\pi}\right)_4 = \left(\frac{\pi}{\bar{\lambda}}\right)_4.$$

Hence

$$\left(\frac{5}{\pi}\right)_4 = \left(\frac{\pi}{\lambda}\right)_4 \left(\frac{\pi}{\bar{\lambda}}\right)_4.$$

Under the isomorphism $\mathbb{Z}[i]/(\lambda) \simeq \mathbb{F}_5$ one has $i \mapsto 3$, while under $\mathbb{Z}[i]/(\bar{\lambda}) \simeq \mathbb{F}_5$ one has $i \mapsto 2 = 3^{-1}$. Let ℓ denote the discrete logarithm to base 3 on $\mathbb{F}_5^{\times}$. Modulo λ,

$$\pi = a^* + bi \equiv a^*(1 + 3r),$$

and modulo $\bar{\lambda}$,

$$\pi = a^* + bi \equiv a^*(1 + 2r).$$

The factor a^* cancels in the product. Indeed, writing $a^* \equiv 3^k \pmod 5$, its contribution modulo λ is i^k (under $i \mapsto 3$), while its contribution modulo $\bar{\lambda}$ is i^{-k} (under $i \mapsto 3^{-1}$); the product of these contributions is 1. Therefore

$$\left(\frac{5}{\pi}\right)_4 = i^{\,\ell((1+3r)/(1+2r))}.$$

If $r = 1$, then

$$\frac{1+3r}{1+2r} = \frac{4}{3} \equiv 3 \pmod 5,$$

so the exponent is 1 and the symbol is i. If $r = -1$, then

$$\frac{1+3r}{1+2r} = \frac{-2}{-1} \equiv 2 \pmod 5,$$

so the exponent is 3 and the symbol is $i^3 = -i$. This proves the claim. □

Proof of Theorem B. By Lemma 3.2, $p \equiv 1 \pmod 4$ and $z(p) = (p+1)/2$. If $p \equiv 5 \pmod 8$, then $S(p) = 0$ by Corollary 3.5(a).

Now assume $p \equiv 1 \pmod 8$. Proposition 5.2 gives $S(p) = \frac{1}{2}\mathcal{J}(5)$. Since $\varepsilon_p = -1$, 5 is a quadratic non-residue modulo p. Choose any nonzero quadratic residue $r \bmod p$. By Lemma 2.4, $\frac{1}{2}\mathcal{J}(r)$ is odd, $\frac{1}{2}\mathcal{J}(5)$ is even, and

$$\left(\tfrac{1}{2}\mathcal{J}(r)\right)^2 + \left(\tfrac{1}{2}\mathcal{J}(5)\right)^2 = p.$$

The odd component of a representation of p as a sum of two squares is $|a^*|$, and the even component is b. Hence $|\frac{1}{2}\mathcal{J}(5)| = b$.

It remains to determine the sign. By Lemma 2.5,

$$\tfrac{1}{2}\mathcal{J}(5) \equiv -a^*\, 5^{(p-1)/4} \pmod p.$$

Let

$$\pi = a^* + bi.$$

Under the isomorphism

$$\mathbb{Z}[i]/(\pi) \simeq \mathbb{F}_p$$

one has

$$i \equiv -\frac{a^*}{b} \pmod \pi.$$

Moreover,

$$5^{(p-1)/4} \equiv \left(\frac{5}{\pi}\right)_4 \quad \text{in } \mathbb{Z}[i]/(\pi).$$

If $b \equiv a^* \pmod 5$, Lemma 5.4 gives

$$\left(\frac{5}{\pi}\right)_4 = i.$$

Hence, in $\mathbb{F}_p$,

$$5^{(p-1)/4} \equiv -\frac{a^*}{b}.$$

Therefore

$$\tfrac{1}{2}\mathcal{J}(5) \equiv -a^*\Big(-\frac{a^*}{b}\Big) = \frac{(a^*)^2}{b} \equiv -b \pmod p,$$

because $(a^*)^2 + b^2 = p \equiv 0 \pmod p$.

If $b \equiv -a^* \pmod 5$, Lemma 5.4 gives

$$\left(\frac{5}{\pi}\right)_4 = -i.$$

Hence

$$5^{(p-1)/4} \equiv \frac{a^*}{b} \pmod p,$$

and therefore

$$\tfrac{1}{2}\mathcal{J}(5) \equiv -a^*\frac{a^*}{b} = -\frac{(a^*)^2}{b} \equiv b \pmod p.$$

Since $\frac{1}{2}\mathcal{J}(5) = \pm b$ and $b \not\equiv -b \pmod p$, these congruences determine the sign:

$$\tfrac{1}{2}\mathcal{J}(5) = \begin{cases} -b, & b \equiv a^* \pmod 5, \\ b, & b \equiv -a^* \pmod 5. \end{cases}$$

Equivalently, $S(p)$ is the unique element of $\{\pm b\}$ congruent to $-a^*$ modulo 5. □

Remark 5.5. Split index two thus realises the *residue* Jacobsthal component $(\pm a^*)$, and inert index two realises the *non-residue* component $(\pm b)$.

Remark 5.6 (Why parameter 5 rather than 10)**.** For $p \equiv 1 \pmod 8$ write $2 = s^2$. Then $\mathcal{J}(10) = \mathcal{J}(5s^2) = \chi(s)\mathcal{J}(5) = \left(\frac{2}{p}\right)_4 \mathcal{J}(5)$. Hence $|\mathcal{J}(\pm 10)| = 2|S(p)|$, but the sign is twisted by the quartic symbol $\left(\frac{2}{p}\right)_4$.

6. Numerical evidence

The companion verifier `ghandali_verify_index2.py` (version 1.0.2), archived with its machine-readable outputs and run log in [6], was run with

$$N = 2 \cdot 10^6, \qquad \texttt{sum_limit} = 2 \cdot 10^6, \qquad \texttt{jac_limit} = 2 \cdot 10^6,$$

computing direct Fibonacci character sums for rank indices 2 and 4. The run formed a rank inventory of 148,931 primes $p \le N$ with $p \nmid 10$; all theorem-level cases reported in the table below satisfy the standing hypothesis $p > 5$. It selected 52,477 direct $S(p)$-records and returned zero mismatches in the theorem-level identities checked by the verifier.

Case	Prediction	Primes	Agreement	Proved in
inert, $i_p = 2$, $p \equiv 1 \pmod 8$	$S \in \{\pm b\}$, $S \equiv -a^* \pmod 5$	14591	14591/14591	Thm. B
inert, $i_p = 2$, $p \equiv 5 \pmod 8$	$S = 0$	14696	14696/14696	Cor. 3.5
split, $i_p = 2$, $p \equiv 1 \pmod 8$	$S = \delta_p a^*$	6551	6551/6551	Thm. A
split, $i_p = 2$, $p \equiv 5 \pmod 8$	$S = -\delta_p$	6583	6583/6583	Thm. A
split, $i_p = 4$, $p \equiv 5 \pmod 8$	$S = 0$	6697	6697/6697	Cor. 3.5
split, $i_p = 4$, $p \equiv 1 \pmod 8$	S odd; exact value open	3359	3359/3359	Lem. 3.1

The same run also checked the rank identity $z(p)i_p = n_p$ and the index-parity criterion on all 148,931 primes in the rank inventory. The identity

$$\chi(F_{z(p)+1}) = \left(\frac{2}{p}\right)$$

for odd $z(p)$ was checked on 49,618 primes, and the reflection criteria used in Corollary 3.5 were checked with zero mismatches. The split and inert Jacobsthal reductions were checked directly throughout the same range wherever the corresponding direct Jacobsthal sums were computed; in particular,

$$2S(p) = \mathcal{J}(5)$$

was checked for all 14,591 inert index-two primes with $p \equiv 1 \pmod 8$ below $2 \cdot 10^6$. The same run verified the signed law $S(p) \equiv -a^* \pmod 5$ on those primes with zero mismatches.

Claims of zero mismatches refer to the archived run deposited in [6], which contains the verifier source code (version 1.0.2), machine-readable CSV and JSON output files, the run log with the parameters recorded above, and checksums. The present paper records only the mathematical summary of these computations.

7. Conditional density context

This section is included only as conditional context under the generalized Riemann hypothesis and is not used in the proofs of Theorems A and B. Those theorems are unconditional.

7.1. **A conditional density corollary for inert index two.** The computations in Section 6 suggest the two index-two density constants

$$\frac{9A}{38} \qquad \text{in the split branch}$$

and

$$\frac{10A}{19} \qquad \text{in the inert branch,}$$

where A denotes Artin's constant. The split value is recorded here as a conjectural density suggested by the data. In contrast, the inert value follows, as a Dirichlet density, conditionally from the GRH-dependent density theorems of Roskam [20] and Chen [2], in the general Artin-type framework of Lenstra [14], once their maximal-order statements are translated into the present rank-index notation; the derivation is self-contained in Corollary 7.2 and does not rely on [5].
The rank index as a group index. Let $p \nmid 10$, and write

$$\eta = \frac{\varphi}{\psi} = -\varphi^2.$$

If $\left(\frac{5}{p}\right) = 1$, put $G_p = \mathbb{F}_p^\times$. If $\left(\frac{5}{p}\right) = -1$, let

$$G_p = T(\mathbb{F}_p) = \ker\Big(N : \mathbb{F}_{p^2}^\times \to \mathbb{F}_p^\times\Big)$$

be the norm-one torus. In both cases G_p is cyclic of order

$$n_p = p - \left(\frac{5}{p}\right),$$

and $\eta \in G_p$. Since

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}},$$

one has $F_n \equiv 0 \pmod{p}$ if and only if $\eta^n = 1$. Hence

$$z(p) = \mathrm{ord}_{G_p}(\eta), \qquad i_p = [G_p : \langle \eta \rangle].$$

Thus full rank $i_p = 1$ means that η is a primitive point of the one-dimensional torus attached to $\mathbb{Q}(\sqrt{5})$, while rank index t means that η generates the unique subgroup of index t. This is the torus-theoretic form of the generalized Artin problem in the sense of Lenstra [14], and is the setting of Chen's theorem on primitive points of one-dimensional tori [2].

Lemma 7.1 (2-adic dictionary, inert branch)**.** *Let $p \neq 2, 5$ be inert in $\mathbb{Q}(\sqrt{5})$, and put*

$$d = \mathrm{ord}(\bar{\varphi}^{\,2}),$$

where $\bar{\varphi}$ denotes the image of φ in $\mathbb{F}_{p^2}^{\times}$. Then:

(i)
$$\bar{\varphi}^{\,p+1} = N(\bar{\varphi}) = -1.$$
Consequently $d \mid p+1$, the quotient $(p+1)/d$ is odd, and
$$\bar{\eta} = (\bar{\varphi}^{\,2})^{(p+3)/2}.$$

(ii) *If $p \equiv 3 \pmod 4$, then*
$$\gcd\left(d, \frac{p+3}{2}\right) = 1,$$
so
$$z(p) = d, \qquad i_p = \frac{p+1}{d},$$
and i_p is odd.

(iii) *If $p \equiv 1 \pmod 4$, then $d = 2u$ with u odd and*
$$\gcd\left(d, \frac{p+3}{2}\right) = 2.$$
Therefore
$$z(p) = u = \frac{d}{2}, \qquad i_p = \frac{p+1}{u} = \frac{2(p+1)}{d} \equiv 2 \pmod 4.$$

In particular,

$$\{\, p \text{ inert} : \ i_p \in \{1, 2\} \,\} = \{\, p \text{ inert} : \ d = p+1 \,\}. \tag{1}$$

The set on the right is the inert maximal-order set for the fundamental unit φ, equivalently the set denoted S^- in Roskam's specialization to $\mathbb{Q}(\sqrt{5})$ [20].

Proof. Since p is inert in $\mathbb{Q}(\sqrt{5})$, Frobenius interchanges $\bar{\varphi}$ and $\bar{\psi}$. Hence

$$\bar{\varphi}^{\,p+1} = \bar{\varphi}\,\bar{\varphi}^{\,p} = \bar{\varphi}\,\bar{\psi} = -1.$$

Let $m = \mathrm{ord}(\bar{\varphi})$. Then $m \mid 2(p+1)$, while $m \nmid p+1$, because $\bar{\varphi}^{\,p+1} = -1$. Therefore

$$d = \mathrm{ord}(\bar{\varphi}^{\,2})$$

divides $p+1$, but $d \nmid (p+1)/2$. Equivalently, $(p+1)/d$ is odd. Moreover

$$-1 = \bar{\varphi}^{\,p+1} = (\bar{\varphi}^{\,2})^{(p+1)/2},$$

and hence

$$\bar{\eta} = -\bar{\varphi}^{\,2} = (\bar{\varphi}^{\,2})^{(p+3)/2}.$$

This proves (i).

For (ii), write $p = 4k+3$. Then $(p+3)/2 = 2k+3$ is odd. Any common divisor of d and $(p+3)/2$ divides

$$2 \cdot \frac{p+3}{2} - (p+1) = 2.$$

Since $(p+3)/2$ is odd, this common divisor must be 1. Therefore

$$z(p) = \operatorname{ord}(\bar{\eta}) = \frac{d}{\gcd(d, (p+3)/2)} = d,$$

and so

$$i_p = \frac{p+1}{z(p)} = \frac{p+1}{d},$$

which is odd by (i).

For (iii), write $p = 4k+1$. Then $v_2(p+1) = 1$. Since $(p+1)/d$ is odd, we have $v_2(d) = 1$, so $d = 2u$ with u odd. Also

$$\frac{p+3}{2} = \frac{p+1}{2} + 1$$

is even, and

$$\gcd\left(u, \frac{p+3}{2}\right) \mid \gcd\left(\frac{p+1}{2}, \frac{p+3}{2}\right) = 1.$$

Thus

$$\gcd\left(d, \frac{p+3}{2}\right) = 2.$$

It follows that

$$z(p) = \frac{d}{2} = u, \qquad i_p = \frac{p+1}{u} = \frac{2(p+1)}{d}.$$

Since $(p+1)/d$ is odd, this gives $i_p \equiv 2 \pmod 4$.

Finally, if $p \equiv 3 \pmod 4$, then i_p is odd and $i_p = 1$ is equivalent to $d = p+1$. If $p \equiv 1 \pmod 4$, then $i_p \equiv 2 \pmod 4$, and $i_p = 2$ is again equivalent to $d = p+1$. This proves (1). The final identification with Roskam's inert maximal-order set is exactly the condition that $\bar{\varphi}^{\,2}$ generate the norm-one group of order $p+1$, as in Roskam's quadratic analogue of Artin's conjecture [20]. □

Corollary 7.2 (GRH; Roskam–Chen)**.** *Assume the generalized Riemann hypothesis for the Kummer extensions occurring in the density theorems of Roskam [20] and Chen [2]. Then the set of primes $p > 5$ such that*

$$\left(\frac{5}{p}\right) = -1, \qquad i_p = 2$$

has Dirichlet density

$$\frac{10A}{19}$$

among all rational primes, where A denotes Artin's constant. This is a specialization of the cited conditional density theorems into the present rank-index notation; it is not an independent density theorem of this paper and is not used in the proofs of Theorems A and B.

Proof. By Lemma 7.1, the inert primes with $i_p \in \{1, 2\}$ are precisely the inert primes for which

$$\operatorname{ord}(\bar{\varphi}^{\,2}) = p+1.$$

This is Roskam's inert maximal-order set S^- for the fundamental unit φ of $\mathbb{Q}(\sqrt{5})$. Since $N(\varphi) = -1$ and $D = 5$ is odd, Roskam's GRH-conditional Theorem 2(b) [20] gives the *natural* density

$$\delta_{\mathrm{nat}}(S^-) = A\big(1 - E(5)\big) = A\left(1 + \frac{1}{19}\right) = \frac{20A}{19},$$

where, for square-free a,

$$E(a) = \mu(a) \prod_{q|a} \frac{1}{q(q-1)-1}, \qquad E(5) = -\frac{1}{5 \cdot 4 - 1} = -\frac{1}{19}.$$

The erratum to [20] corrects Lemmas 6–7 and the formulas of Theorem 2(a) for the case $N(\varepsilon) = +1$ and for $\delta(S^+)$ when $2 \mid D$; it does not alter the formula $\delta(S^-) = A(1 - E(D))$ used here for $N(\varepsilon) = -1$. A set with natural density has the same Dirichlet density, so

$$\delta_D(S^-) = \frac{20A}{19}.$$

We now compute the Dirichlet density of the inert full-rank set directly from Chen's Theorem 2.1 [2], applied to

$$\eta = -\varphi^2 \in K = \mathbb{Q}(\sqrt{5}).$$

The element η is nontorsion with $N_{K/\mathbb{Q}}(\eta) = 1$ and is not a proper power in $K^\times$: it is negative under both real embeddings of K, so it has no square root in the real field K; any β with $\beta^q = \eta$ is a unit of K, hence of the form $\pm\varphi^m$; and an equality $\eta = (\pm\varphi^m)^q$ with q an odd prime would force $mq = 2$, which is impossible. Hence the exponent s in Chen's notation equals 1.

For square-free n, Chen's field k_n equals K if n is odd and $K(\sqrt{\eta}) = K(i)$ if n is even. Consequently the intersection degree

$$[\, k_n \cap \mathbb{Q}(\mu_n) : \mathbb{Q} \,] = \begin{cases} 2, & 5 \mid n, \\ 1, & 5 \nmid n, \end{cases}$$

because $K = \mathbb{Q}(\sqrt{5})$ is contained in $\mathbb{Q}(\mu_n)$ exactly when $5 \mid n$, while a square-free n is never divisible by 4, so $\mathbb{Q}(\mu_n)$ cannot contain i. Equivalently, the quadratic subfields of $K(i)$ are $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(i)$, and $\mathbb{Q}(\sqrt{-5})$, of conductors 5, 4, and 20, and only 5 can divide a square-free n. Chen's Lemma 1.6 [2] therefore gives the Kummer degrees

$$d_n = [E_n : \mathbb{Q}] = \begin{cases} n\varphi(n), & 5 \mid n, \\ 2n\varphi(n), & 5 \nmid n. \end{cases}$$

For the inert obstruction coefficients, Chen's Lemma 1.7(a) [2] gives $c_n^- = 0$ when $5 \mid n$, since K is real and contained in $\mathbb{Q}(\mu_n)$; and $c_n^- = 1$ when $5 \nmid n$, since then none of the three obstruction conditions of that lemma applies: K is not contained in $\mathbb{Q}(\mu_n)$, the field $K' = \mathbb{Q}(i)$ is not contained in $\mathbb{Q}(\mu_n)$ for square-free n, and $K'' = \mathbb{Q}(\sqrt{-5})$ is imaginary. Chen's formula $\operatorname{den}(M_\eta^-) = \sum_n c_n^- \, \mu(n)/d_n$ [2, Theorem 2.1] therefore yields

$$\delta_D(M_\eta^-) = \sum_{5 \nmid n} \frac{\mu(n)}{2n\varphi(n)} = \frac{1}{2} \prod_{q \neq 5} \Big(1 - \frac{1}{q(q-1)}\Big) = \frac{1}{2} \cdot \frac{A}{1 - 1/20} = \frac{10A}{19}.$$

Since $z(p) = \operatorname{ord}(\bar{\eta})$ and, for inert p, $i_p = (p+1)/z(p)$, the set M_η^- is exactly the inert full-rank set $\{p \text{ inert} : i_p = 1\}$.

The sets $\{i_p = 1\}$ and $\{i_p = 2\}$ are disjoint (Lemma 7.1(ii)–(iii): $i_p = 1$ forces $p \equiv 3 \pmod 4$, while $i_p = 2$ forces $p \equiv 1 \pmod 4$), and their union is S^- up to finitely many excluded primes. Dirichlet density is additive for finite disjoint decompositions, so

$$\begin{aligned} \delta_D\{\, p > 5 : \left(\frac{5}{p}\right) = -1,\ i_p = 2 \,\} &= \delta_D(S^-) - \delta_D(M_\eta^-) \\ &= \frac{20A}{19} - \frac{10A}{19} = \frac{10A}{19}. \end{aligned}$$

□

Remark 7.3 (Natural versus Dirichlet density)**.** The corollary is stated for Dirichlet density, which is what the above argument yields. Roskam's Theorem 2(b) in fact provides the natural density of S^-, but Chen's Theorem 2.1 is stated for Dirichlet density; subtracting a Dirichlet density from a natural density does not, by itself, establish natural density for the difference set. Promoting the statement to natural density would require an effective Hooley-type argument (GRH-conditional Chebotarev estimates with tail control) applied to the inert full-rank set $\{i_p = 1\}$ directly, along the lines of [17]; the convergence of the Euler product in the computation above is not by itself such an estimate. The numerical data of Section 6 is of course consistent with either notion of density. We record the Dirichlet-density statement, which is the strongest that the cited references directly support.

Remark 7.4 (Numerical confirmation and the split branch)**.** The Dirichlet-density value in Corollary 7.2 is also reflected numerically. In the run up to $2 \cdot 10^6$, there are

$$14{,}591 + 14{,}696 = 29{,}287$$

inert primes with $i_p = 2$. Writing $C(N) = 29{,}287$ for this count and letting $\pi^*(N)$ denote the number of primes $p \le N$ with $p \nmid 10$, one has

$$\frac{C(N)}{\pi^*(N)A} \approx 0.525859,$$

compared with the GRH-conditional prediction

$$\frac{10}{19} \approx 0.526316.$$

The relative deviation is about 0.09%.

The split index-two density suggested by the data is

$$\frac{9A}{38}.$$

In the same run, if $C_{\mathrm{split}}(N)$ denotes the number of split index-two primes up to N, one has

$$\frac{C_{\mathrm{split}}(N)}{\pi^*(N)A} \approx 0.235826,$$

compared with

$$\frac{9}{38} \approx 0.236842.$$

Unlike the inert branch, the split index-two set does not appear to admit the same direct dictionary with Roskam's split maximal-order set. The obstruction is the 2-primary Kummer behaviour described in Remark 7.5; see also the discussion of the full-rank split branch in [5]. Thus the split value is recorded here as a conjectural density supported by the computation, not as a theorem. Its eventual proof should fall within the near-primitive-root framework treated in [11, 15, 18].

Remark 7.5 (Kummer degree when $4 \mid n$)**.** Write

$$K_n = \mathbb{Q}(\sqrt{5}, \zeta_n).$$

Density computations of Artin type for the subgroup generated by η depend on the Kummer degrees

$$[K_n(\eta^{1/n}) : K_n].$$

If $4 \mid n$, then $i = \zeta_4 \in K_n$, and

$$\eta = -\varphi^2 = (i\varphi)^2$$

is already a square in K_n. Consequently

$$[K_n(\eta^{1/n}) : K_n] \le \frac{n}{2}.$$

Thus any density computation which treats this degree as n without correction is valid only away from the $4 \mid n$ levels. This 2-primary obstruction is precisely why the split index-two density requires a separate near-primitive-root analysis, rather than a naive subtraction from Roskam's split maximal-order set. Compare the general Artin-type frameworks of Lenstra [14] and Chen [2].

Numerical comparison. Table 1 records the aggregate density ratios obtained from the $N = 2 \cdot 10^6$ verification run. The density rows are included as context for the rank-index distribution; the finite computation is not used as a proof of the GRH-conditional density Corollary 7.2.

Quantity	Unit	Prediction	$N = 2 \cdot 10^6$
full rank, all	A	1	1.00128
full rank, split	A	$9/19 = 0.47368$	0.47415
full rank, inert	A	$10/19 = 0.52632$	0.52713
$i_p = 2$, split	A	$9/38 = 0.23684$ conjectural	0.23583
$i_p = 2$, inert	A	$10/19 = 0.52632$ Dirichlet, conditional	0.52586

TABLE 1. Aggregate rank-index density ratios from the companion verification run with $N = 2 \cdot 10^6$. The density framework is the GRH-conditional Artin–Lenstra–Chen–Roskam framework [2, 14, 20]. The inert index-two value $10A/19$ is the *Dirichlet* density of Corollary 7.2 (see Remark 7.3). The split index-two value $9A/38$ is recorded only as a conjectural density suggested by the data. The finite computation is not used as a proof of any density statement.

7.2. **Framework and the series.**

Remark 7.6 (Framework for index densities)**.** The determination of densities at fixed rank index is an instance of the generalized Artin problem in the sense of Lenstra [14]. Murty [17] supplies a conditional analytic framework for Artin-type problems. The quadratic-field case for units is treated by Ishikawa–Kitaoka [8], Chen–Kitaoka–Yu [3], Roskam [19, 20], and Kataoka [10, 11]. Near-primitive-root and character-sum formulations of primitive-root densities are treated in [15, 18]. In the present setting the relevant point is the norm-one element

$$\eta = -\varphi^2$$

on the torus attached to $\mathbb{Q}(\sqrt{5})$. The full-rank case $i_p = 1$ is governed, under GRH, by Chen's density theorem for primitive points on one-dimensional tori [2], while the inert index-one/index-two union is identified in Lemma 7.1 with Roskam's inert maximal-order set for the fundamental unit φ. The present paper uses this density framework only for contextual information; the proofs of Theorems A and B are independent of GRH.

Remark 7.7 (Relation to the author's series)**.** The present paper is part of a sequence of works on rank-block decompositions and character sums in Fibonacci and Lucas sequences. The paper on half-period involutions [4] treats exact cancellation mechanisms in inert Lucas sequences, and the full-rank paper [5] evaluates $S(p)$ at $i_p = 1$. The present paper contributes the rank-index-two reductions

$$S(p) = \frac{1}{2}\mathcal{J}(\cdot)$$

together with the resulting explicit evaluations in Theorems A and B. Rank-multiplier and reflection identities used here are classical in origin, going back to Lucas [16] and later treatments of Lucas sequences such as Somer [21]; see also [4]. No proof in the present paper depends on an unpublished manuscript.

8. Further problems and directions

Remark 8.1 (Signed inert two-square law)**.** The sign in the inert case $i_p = 2$, $p \equiv 1 \pmod 8$ is not an open refinement: Theorem B determines it by the congruence $S(p) \equiv -a^* \pmod 5$. The companion verifier (`ghandali_verify_index2.py`, version 1.0.2) checks this signed law directly (zero mismatches on all 14,591 eligible primes below $2 \cdot 10^6$).

Remark 8.2 (Index four: vanishing and parity)**.** For split primes with $i_p = 4$ and $p \equiv 5 \pmod 8$, Corollary 3.5(b) gives $S(p) = 0$ (reflection vanishing). For the complementary split branch $p \equiv 1 \pmod 8$, the exact value of $S(p)$ remains open, but its parity is settled: here $z(p) = (p-1)/4$ is even, so Lemma 3.1 yields

$$S(p) \equiv z(p) - 1 \equiv 1 \pmod 2.$$

Thus $S(p)$ is odd for every such prime. In the archived computation up to $2 \cdot 10^6$ this oddness held with zero mismatches on 3359 primes, and the observed ratio $|S(p)|/\sqrt{p}$ stayed below 0.503. A natural next step is an analogue of Proposition 4.1 involving quartic Jacobsthal or Jacobi sums that would determine the exact value.

Problem 8.3 (Index three)**.** For rank index three, a precise analogue of the index-two reduction to Jacobsthal sums is not known. A natural hypothesis is that the governing objects are cubic character sums (or cubic Jacobi sums), but no such identity is claimed or archived in the present paper.

Problem 8.4 (Other Lucas sequences)**.** The reductions in Propositions 4.1 and 5.2 use the identities

$$\varphi\psi = -1$$

and the associated Cayley parametrisation. They therefore extend formally to Lucas sequences $U_n(P, -1)$, for which the discriminant 5 is replaced by $P^2 + 4$. The final closed evaluations in Theorem A, however, rely on the special quartic character identity for the Fibonacci unit. A complete analogue for general P would require a corresponding evaluation of the quartic character of the relevant fundamental unit.

Problem 8.5 (Arbitrary fixed index)**.** For a general fixed index $t \geq 1$, can the character sum

$$S(p) = \sum_{j=1}^{z(p)} \chi(F_j)$$

be expressed systematically in terms of Jacobi sums or Jacobsthal sums of order t or $2t$? The full-rank case $t = 1$ and the present rank-index-two case suggest that fixed-index Fibonacci character sums should admit a uniform description through finite-field character sums.

Declarations

No external funding was received for this work. The author declares no competing interests.
Data availability. The companion verifier `ghandali_verify_index2.py` (version 1.0.2), machine-readable outputs, run log, and checksum manifest are available in the reproducibility archive [6] (DOI: 10.5281/zenodo.23263042)

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